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Module: CI5250 Computing Systems

Coursework: Networking Design

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Workshop Tasks Completed:

Please complete the table below.

Week	Tasks Completed (YES/NO)	Additional Remarks
Task 1	Yes	
Task 2	Yes	
Task 3	Yes	
Task 4	Yes	
Task 5	Yes	Task 6 – Email Server

Have you completed the Declaration for the Usage of GAI?

The last page of this template is **COMPULSORY** for everyone, you must fill the form provided there, without doing that, your assignment will **NOT** be marked.

UNDERTAKING:

- By providing my name, KU ID and/or submitting this report on CANVAS, I fully acknowledge that I am aware of the 10% marks for presentation of this report.
- I have fully read the assessment brief, and I understand each and every aspect of the submission guidelines.
- I also fully acknowledge that if I don't submit my packet tracer file separately on CANVAS, my report will not be marked.

Task 1

1. What is the IP address you get after replacing X and Y? (Double check it and leave a sentence mentioning which numbers you have used from your ID)

200.142.199.0/25

I used the 2nd and 4th digits of my K-number (K2432045).

2. Using bullet points (max 5), give step-by-step details of how you performed subnetting for your IP address. These details should not be vague and meaningless, and should show your knowledge of networking, specifically subnetting (Word limit: 100).

- From the network address provided, find the subnet mask, network bits and the host bits:
 - Network address 200.142.199.0/25
 - Subnet mask: 255.255.255.128 – 1111 1111. 1111 1111. 1111 1111. 1000 0000
 - Network bits: 25
 - Host bits: 7
- Identify how many subnets are needed and calculate the new subnet mask and prefix:
 - 4 subnets will be required, which means we need to borrow 2 bits from the host bits.
 - Original subnet mask 255.255.255.128 – Long format 1111 1111. 1111 1111. 1111 1111. 1000 0000
 - New subnet mask 255.255.255.224 – Long format 1111 1111. 1111 1111. 1111 1111. 1110 0000
 - Old prefix 25, new prefix 27
- With the network address and the new subnet mask, calculate each subnet ID, first host, last host and broadcast IP addresses:
 - Each subnet ID will be calculated by replacing the borrowed bits with values
 - Subnet 1 - 200.142.199. 0000 0000 – 200.142.199.0/27
 - Subnet 2 - 200.142.199. 0010 0000 – 200.142.199.32/27
 - Subnet 3 - 200.142.199. 0100 0000 – 200.142.199.64/27
 - Subnet 4 - 200.142.199. 0110 0000 – 200.142.199.96/27
 - Each subnet first host is calculated by replacing the last bit of the host bits with 1
 - Example of Subnet 1: 200.142.199. 0000 0001
 - Each subnet last host is calculated by replacing all host bits with 1 and deducting 1
 - Example of Subnet 1: 200.142.199. 0001 1110
 - Each subnet broadcast channel is calculated by replacing all host bits with 1
 - Example of Subnet 1: 200.142.199. 0001 1111
- Router's IP configuration for the subnets is on TASK 3.

3. After performing subnetting, fill in the table below.

Subnet Name	Subnet ID	First Host	Last Host	Broadcast	Subnet Mask
Network 1	200.142.199.0	200.142.199.1	200.142.199.30	200.142.199.31	255.255.255.224
Network 2	200.142.199.32	200.142.199.33	200.142.199.62	200.142.199.63	255.255.255.224
Network 3	200.142.199.64	200.142.199.65	200.142.199.94	200.142.199.95	255.255.255.224
Network 4	200.142.199.96	200.142.199.97	200.142.199.126	200.142.199.127	255.255.255.224

Task 2

1. Using bullet points (max 5), give step-by-step details of how you implemented the 4 DHCP pools for the 4 subnets. Use the knowledge you gained from the lectures, do not provide any vague answers. (Word limit: 100)

1. IP configuration on the DHCP Server:

- Click on the DHCP Server > Desktop > IP Configuration, set IP configuration to static, input the IP (200.142.199.66), subnet mask (255.255.255.224), default gateway of the subnet (200.142.199.65) and the address of the DNS Server (200.142.199.67).

2. Enable the DHCP service on the DHCP Server:

- On the DHCP Server > Services > DHCP, activate the service by pressing “On”.

3. Add DHCP Server pools for the 4 subnets:

- On the DHCP service page, input the name of the subnet pool to use, the default gateway for that subnet, the DNS Server IP address, the start IP address for that subnet, the Subnet Mask and the maximum number of users.
- Router configuration of DHCP Pools on TASK 3.

4. Provide evidence for the 4 DHCP Pools (Readable SNAPSHOTS). Use the captions as follows for each of the 4 snapshots.

The screenshot shows the DHCP Server configuration window. The 'Services' tab is selected, and the DHCP service is turned 'On'. The configuration for 'Subnet 1' is shown, including the interface 'FastEthernet0', pool name 'Subnet 1', default gateway '200.142.199.1', DNS server '200.142.199.67', start IP address '200.142.199.2', and subnet mask '255.255.255.224'. Below the configuration fields is a table of DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0	0.0.0.0	0.0.0.0
Subnet 4	200.142.1...	200.142.1...	200.142.1...	255.255.2...	29	0.0.0.0	0.0.0.0
Subnet 3	200.142.1...	200.142.1...	200.142.1...	255.255.2...	27	0.0.0.0	0.0.0.0
Subnet 2	200.142.1...	200.142.1...	200.142.1...	255.255.2...	27	0.0.0.0	0.0.0.0
Subnet 1	200.142.1...	200.142.1...	200.142.1...	255.255.2...	29	0.0.0.0	0.0.0.0

Snapshot 1: Configuration settings for DHCP Pool for Subnet 1

Snapshot 2: Configuration settings for DHCP Pool for Subnet 2

Snapshot 3: Configuration settings for DHCP Pool for Subnet 3

The screenshot shows the 'DHCP Server' configuration window with the 'Services' tab selected. The 'DHCP' service is enabled. The configuration is for 'Subnet 4' on the 'FastEthernet0' interface. The settings are as follows:

Interface	Service
FastEthernet0	On

Pool Name: Subnet 4

Default Gateway: 200.142.199.97

DNS Server: 200.142.199.67

Start IP Address: 200.142.199.98

Subnet Mask: 255.255.255.224

Maximum Number of Users: 29

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add, Save, Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0	0.0.0.0	0.0.0.0
Subnet 4	200.142.1...	200.142.1...	200.142.1...	255.255.2...	29	0.0.0.0	0.0.0.0
Subnet 3	200.142.1...	200.142.1...	200.142.1...	255.255.2...	27	0.0.0.0	0.0.0.0
Subnet 2	200.142.1...	200.142.1...	200.142.1...	255.255.2...	27	0.0.0.0	0.0.0.0
Subnet 1	200.142.1...	200.142.1...	200.142.1...	255.255.2...	29	0.0.0.0	0.0.0.0

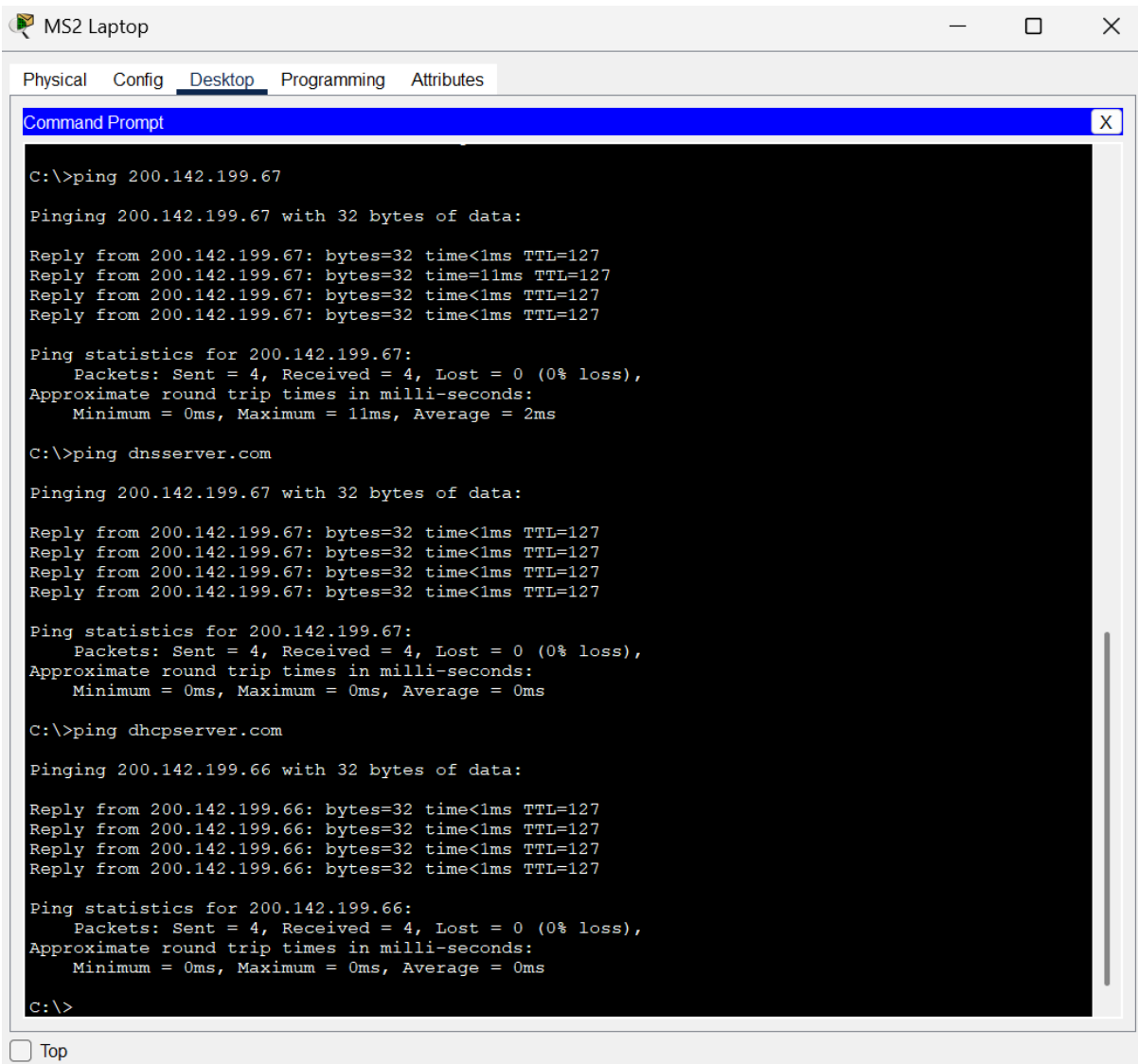
Top

Snapshot 4: Configuration settings for DHCP Pool for Subnet 4

5. Using bullet points (max 3), give step-by-step details of how you implemented the DNS server. Use the knowledge you gained from the lectures, do not provide any vague answers (Word limit: 70)

- IP configuration on the DNS Server:
 - Click on the DNS Server > Desktop > IP Configuration, set IP configuration to static, input the IP (200.142.199.67), subnet mask (255.255.255.224), default gateway of the subnet (200.142.199.65) and the address of the DNS Server (200.142.199.67).
- Enable the DNS service on the DNS Server:
 - On the DNS Server > Services > DNS, activate the service by pressing "On".
- Add "A Record" to the DNS Server:
 - On the DNS service page, insert a name (example: dnsserver.com), choose a type of record (A Record) and input the IP address that you want to refer for this record/name.

4. Provide one evidence for the successful implementation of the DNS Server (**Readable SNAPSHOT**). Replace the following caption based on your evidence.



The screenshot shows a Windows Command Prompt window titled "MS2 Laptop" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt displays the following output:

```
C:\>ping 200.142.199.67

Pinging 200.142.199.67 with 32 bytes of data:

Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time=11ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>ping dnsserver.com

Pinging 200.142.199.67 with 32 bytes of data:

Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping dhcpserver.com

Pinging 200.142.199.66 with 32 bytes of data:

Reply from 200.142.199.66: bytes=32 time<1ms TTL=127
Reply from 200.142.199.66: bytes=32 time<1ms TTL=127
Reply from 200.142.199.66: bytes=32 time<1ms TTL=127
Reply from 200.142.199.66: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Below the Command Prompt window, there is a checkbox labeled "Top" which is currently unchecked.

Snapshot 1: Evidence that the DNS server works properly

Task 3

1. Using bullet points (max 5), give general details of how you performed the device configuration for all the network and end-devices in the 4 subnets. Use the knowledge you gained from the lectures, do not provide any vague answers (Word limit: 100)

1. Laptops, IoT Gateway and IoT Devices configuration:

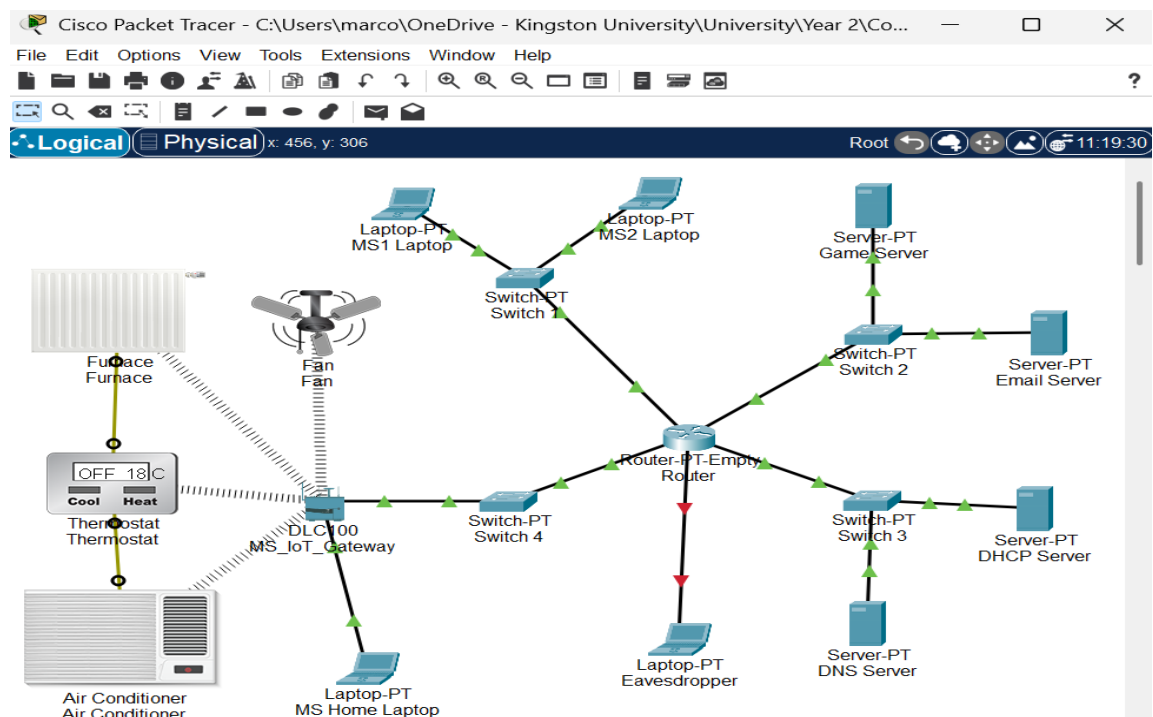
- MS1 and MS2 laptops > Desktop > IP configuration page to set IP attribution to DHCP, which they obtain from the network's DHCP Server.
- MS3 laptop > Desktop > IP configuration page to set IP attribution to DHCP, which obtains its IP from the IoT, using an IP from the IoT > Config > Lan configuration.
- MS_IoT_Gateway > Config > Internet page to set IP attribution to DHCP, which obtains its IP from the network's DHCP Server.
- IoT Devices configuration explained on TASK 4.

2. DHCP, DNS, Email, and Game Servers:

- DHCP and DNS IP configuration explained on task 2.
- Email Server explained on task 6.
- Game Server was given a static IP address (200.142.199.34), subnet mask (255.255.255.224), gateway (200.142.199.33) and DNS server (200.142.199.67) in the IP configuration page.

3. Router IP and DHCP pools configuration:

- Click on the Router > CLI and say no to "continue to configuration dialogue".
- Set a password on the Router by entering privileged mode (enable), entering global configuration mode (configure terminal), then set the password (enable secret "password") replace what's within quotation marks with the password you want.
- Set IP addresses and DHCP pools, while on global configuration mode, enter interface configuration (example: interface g0/0), set the IP address and subnet mask for that interface (example: ip 200.142.199.1 255.255.255.224), make the interface active (no shutdown) and tell the interface about your DHCP pools (ip helper-address 200.142.199.66). Do this for all interfaces.
- Write the executed commands to router's memory by going back to privileged mode and entering the command (wr).



2. Fill in the following table and use the words ‘NOT APPLICABLE’ wherever necessary.

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router (for the 4 subnets)	g0/0	200.142.199.1	255.255.255.224	200.142.199.1
	g1/0	200.142.199.33	255.255.255.224	200.142.199.33
	g2/0	200.142.199.65	255.255.255.224	200.142.199.65
	g3/0	200.142.199.97	255.255.255.224	200.142.199.97
Laptop 1 (subnet 1)	f0	200.142.199.3	255.255.255.224	200.142.199.1
Laptop 2 (subnet 1)	f0	200.142.199.2	255.255.255.224	200.142.199.1
Home Laptop (subnet 4)	f0	192.168.0.106	255.255.255.0	192.168.0.1
DHCP Server	f0	200.142.199.66	255.255.255.224	200.142.199.65
DNS Server	f0	200.142.199.67	255.255.255.224	200.142.199.65
Game Server	f0	200.142.199.34	255.255.255.224	200.142.199.33
Email Server	f0	200.142.199.35	255.255.255.224	200.142.199.33
IoT Gateway	0/0	200.142.199.98	255.255.255.224	200.142.199.97
Thermostat	w0	192.168.0.105	255.255.255.0	192.168.0.1

This IP addresses refers to the IP's assigned at the time of report submission.

3. Provide evidence for the successful connection between difference devices, as follows, using the correct IP addresses.

Pick any device of your choice from the subnets (**Readable SNAPSHOTS**).

IMPORTANT: For Subnet 4, just Ping the IP address of the router interface, as you might not be able to ping the IoT Gateway due to security reasons.

MS2 Laptop

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 42ms, Average = 12ms

C:\>ping 200.142.199.34

Pinging 200.142.199.34 with 32 bytes of data:

Reply from 200.142.199.34: bytes=32 time=16ms TTL=127
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>ping 200.142.199.34

Pinging 200.142.199.34 with 32 bytes of data:

Reply from 200.142.199.34: bytes=32 time<1ms TTL=127
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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Snapshot 1: Successful PING command between Subnet 1 and Subnet 2

MS2 Laptop

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 200.142.199.34: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.142.199.67

Pinging 200.142.199.67 with 32 bytes of data:

Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time=12ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 200.142.199.67

Pinging 200.142.199.67 with 32 bytes of data:

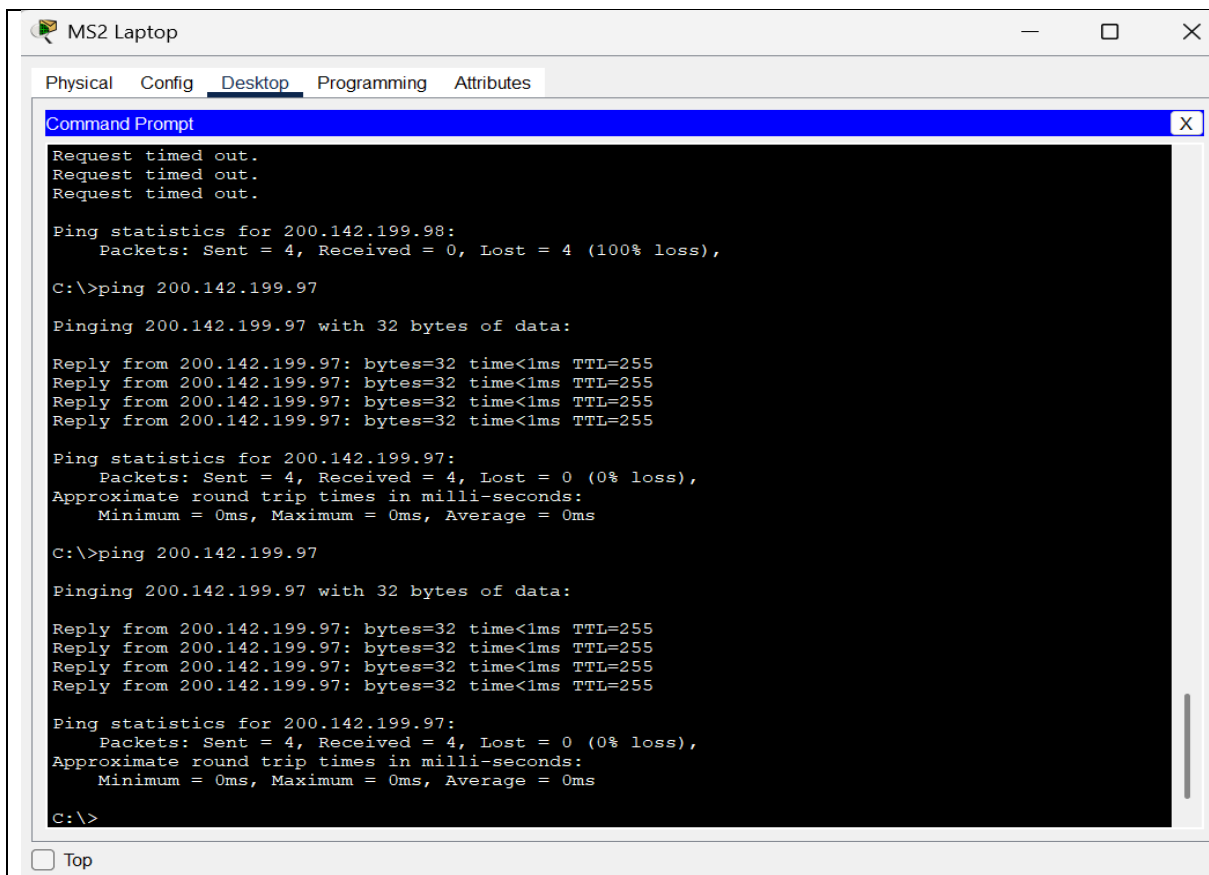
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time=1ms TTL=127
Reply from 200.142.199.67: bytes=32 time=1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

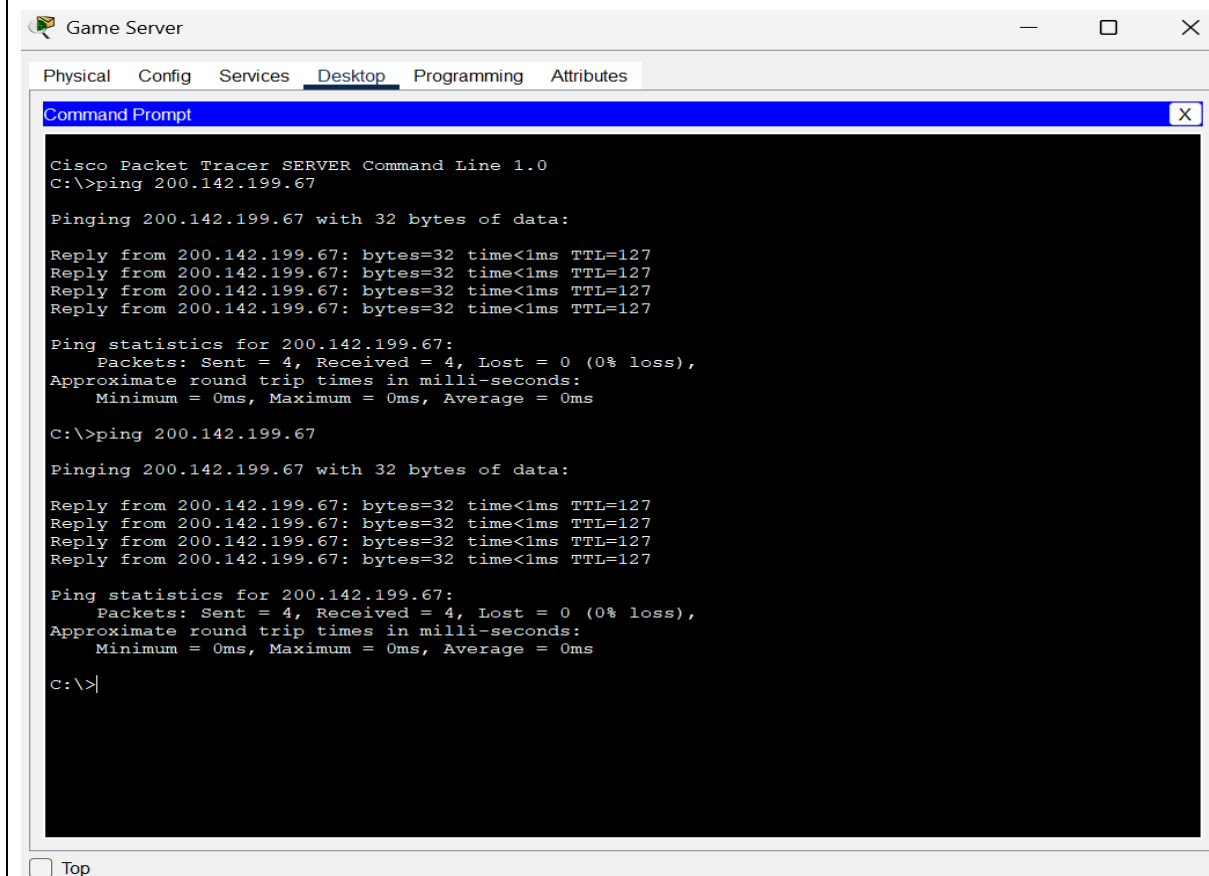
C:\>
```

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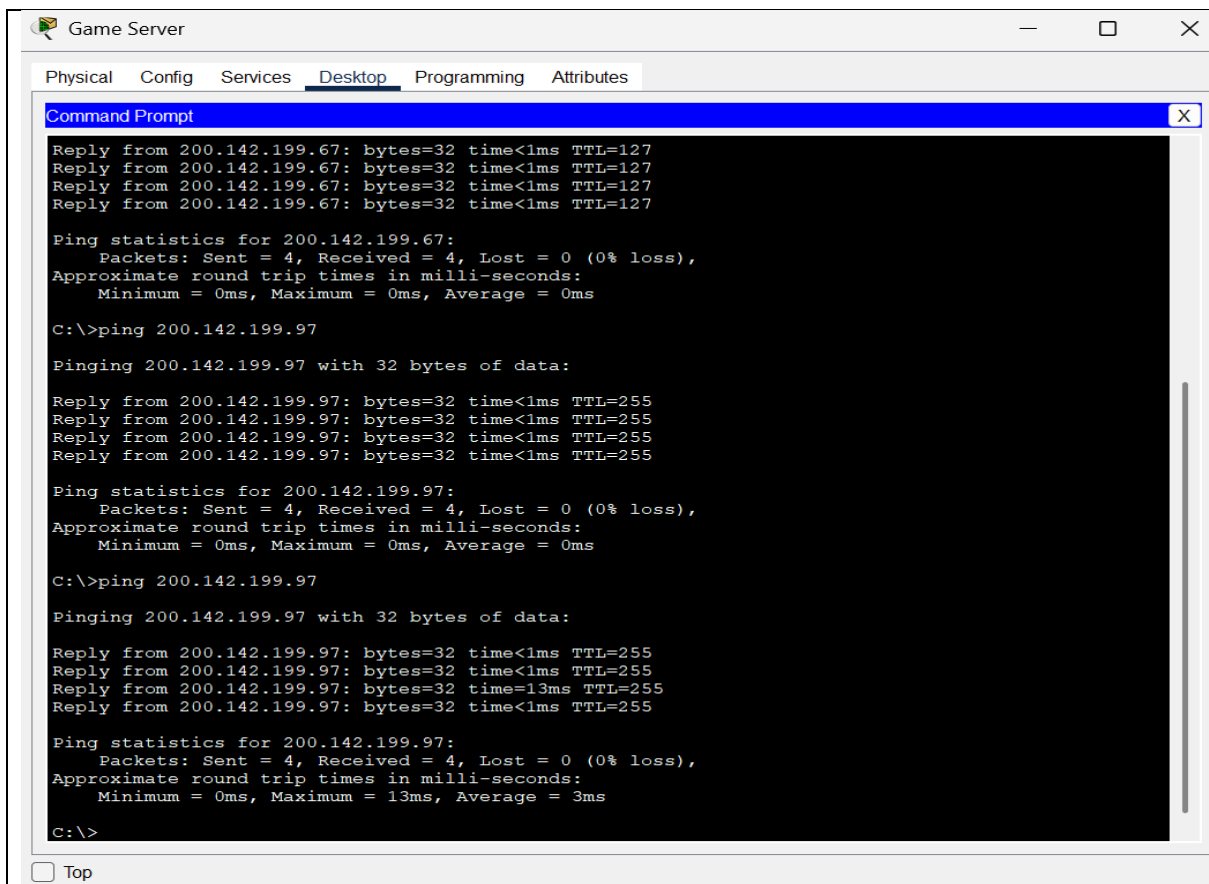
Snapshot 2: Successful PING command between Subnet 1 and Subnet 3



Snapshot 3: Successful PING command between Subnet 1 and Subnet 4



Snapshot 4: Successful PING command between Subnet 2 and Subnet 3



Game Server

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127
Reply from 200.142.199.67: bytes=32 time<1ms TTL=127

Ping statistics for 200.142.199.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.142.199.97

Pinging 200.142.199.97 with 32 bytes of data:

Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255

Ping statistics for 200.142.199.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.142.199.97

Pinging 200.142.199.97 with 32 bytes of data:

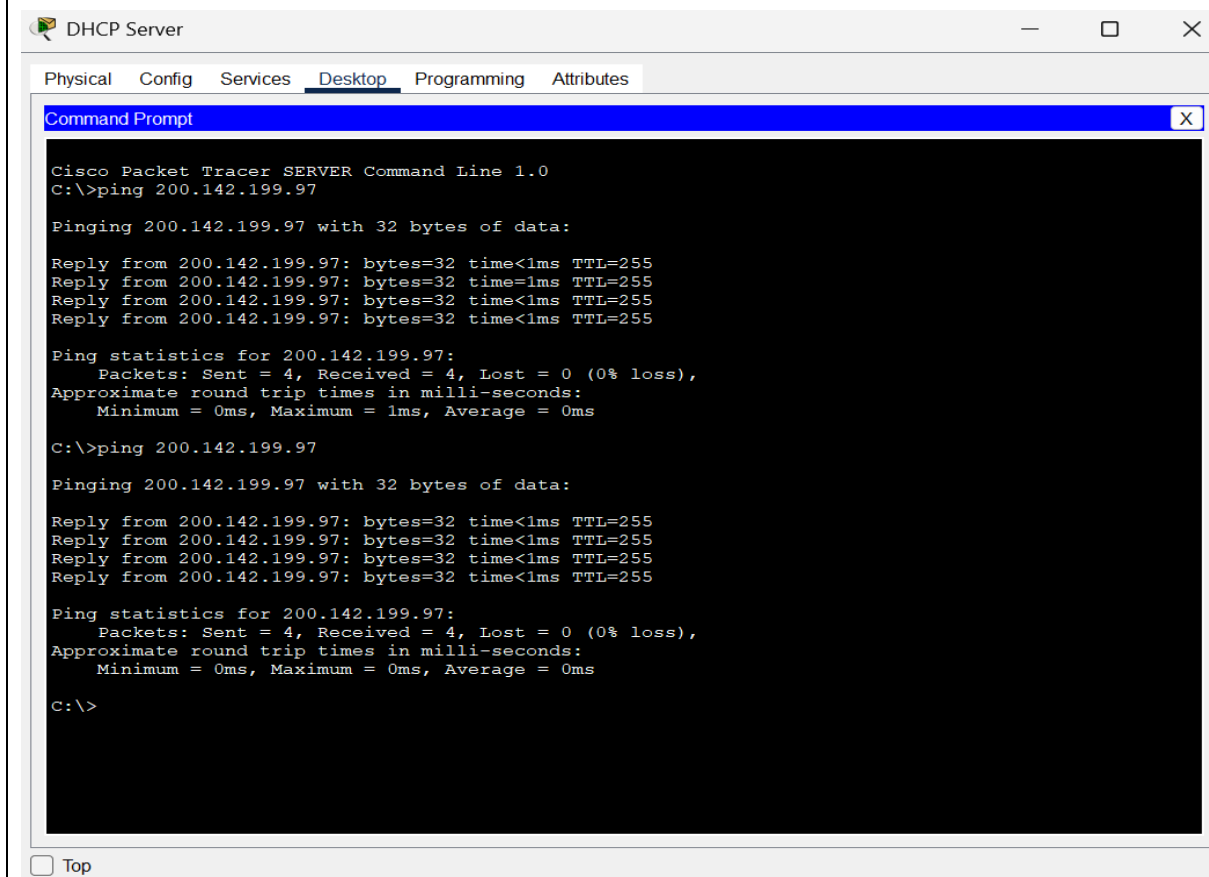
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time=13ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255

Ping statistics for 200.142.199.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>
```

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Snapshot 5: Successful PING command between Subnet 2 and Subnet 4



DHCP Server

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 200.142.199.97

Pinging 200.142.199.97 with 32 bytes of data:

Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255

Ping statistics for 200.142.199.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 200.142.199.97

Pinging 200.142.199.97 with 32 bytes of data:

Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255
Reply from 200.142.199.97: bytes=32 time<1ms TTL=255

Ping statistics for 200.142.199.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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Snapshot 6: Successful PING command between Subnet 3 and Subnet 4

Task 4

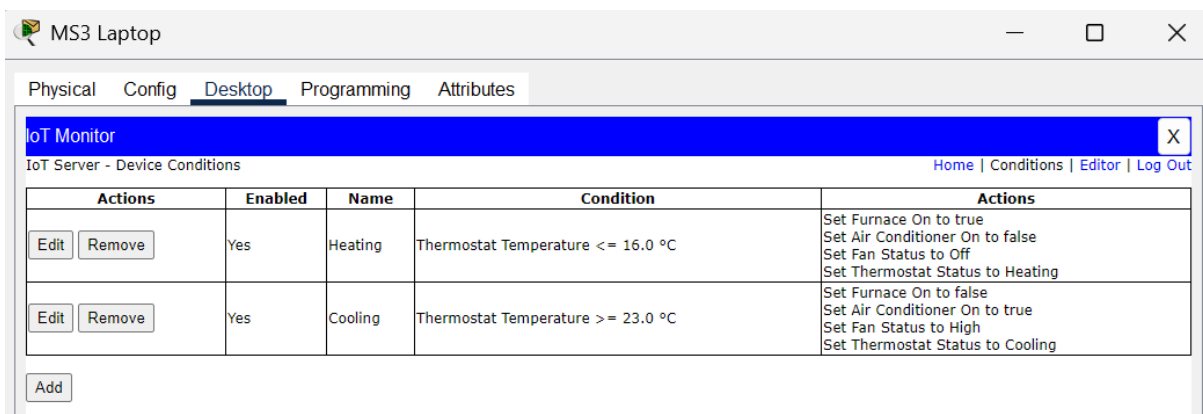
1. Using bullet points (max 6), give details of how you performed the IoT device configuration to connect the devices to the IoT Gateway. Explain your understanding of Smart Homes and IoT. Use the knowledge you gained from the lectures, do not provide any vague answers (Word limit: 150)

1. IP configuration:
 - IoT Devices already come pre-configured to get their IP address by DHCP from the IoT Gateway, just confirm it by clicking on the IoT Device > Config > Settings and check if IPv4 is set to DHCP.
2. IoT Server:
 - Click on the IoT Device > Config > Settings > IoT Server to change to Home Gateway option, allowing this device to be accessed and configured by an IoT Monitor.

2. Using bullet points (max 3), give brief details of how you connected the thermostat with the furnace and the air conditioner. Use the knowledge you gained from the lectures, do not provide any vague answers (Word limit: 75)

1. Physical connection:
 - Use a IoT Custom Cable to connect the furnace (D0) to the thermostat (D1), and connect the air conditioner (D0) to the thermostat (D2).
2. Set up the conditions for the temperature:
 - Click on the laptop connected to the IoT Gateway MS3 Laptop > Desktop > IoT Monitor, enter IoT server address from the laptop perspective (192.168.0.1) and use the username and password already there, go to Conditions and add the following conditions:
 - Name this condition heating, add an if condition thermostat temperature smaller or equal to 16degrees Celsius then furnace on is true, air conditioner on is false, fan status to off and thermostat status to heating.
 - Name this condition cooling, add an if condition thermostat temperature greater or equal to 23degrees Celsius then furnace on is false, air conditioner on is true, fan status to high and thermostat status to cooling.

3. Provide evidence for the successful implementation of the rules for the Smart Home System. (Readable SNAPSHOTS)

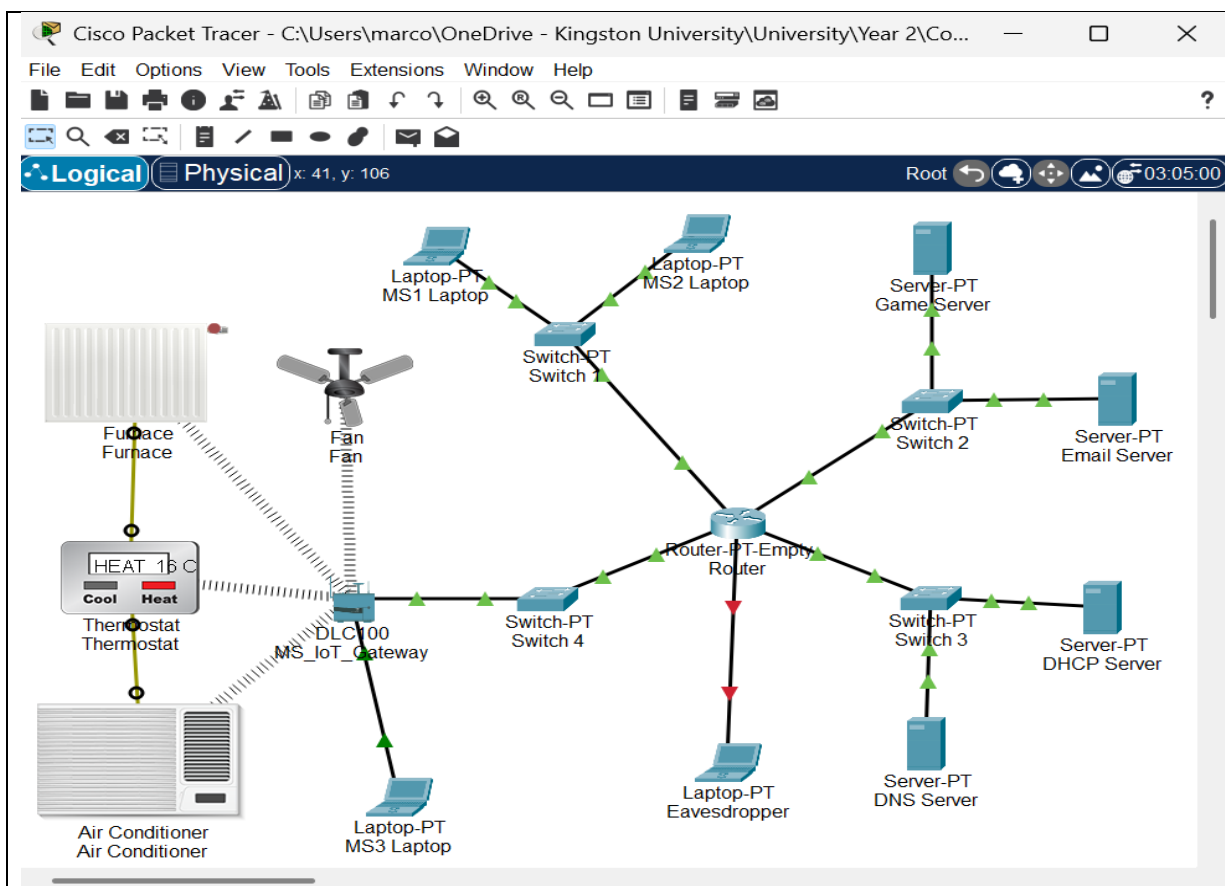


The screenshot shows the 'IoT Monitor' window on an 'MS3 Laptop'. The 'Desktop' tab is selected, displaying 'IoT Server - Device Conditions'. A table lists two conditions: 'Heating' and 'Cooling'. Each condition has a set of actions defined for when the condition is met.

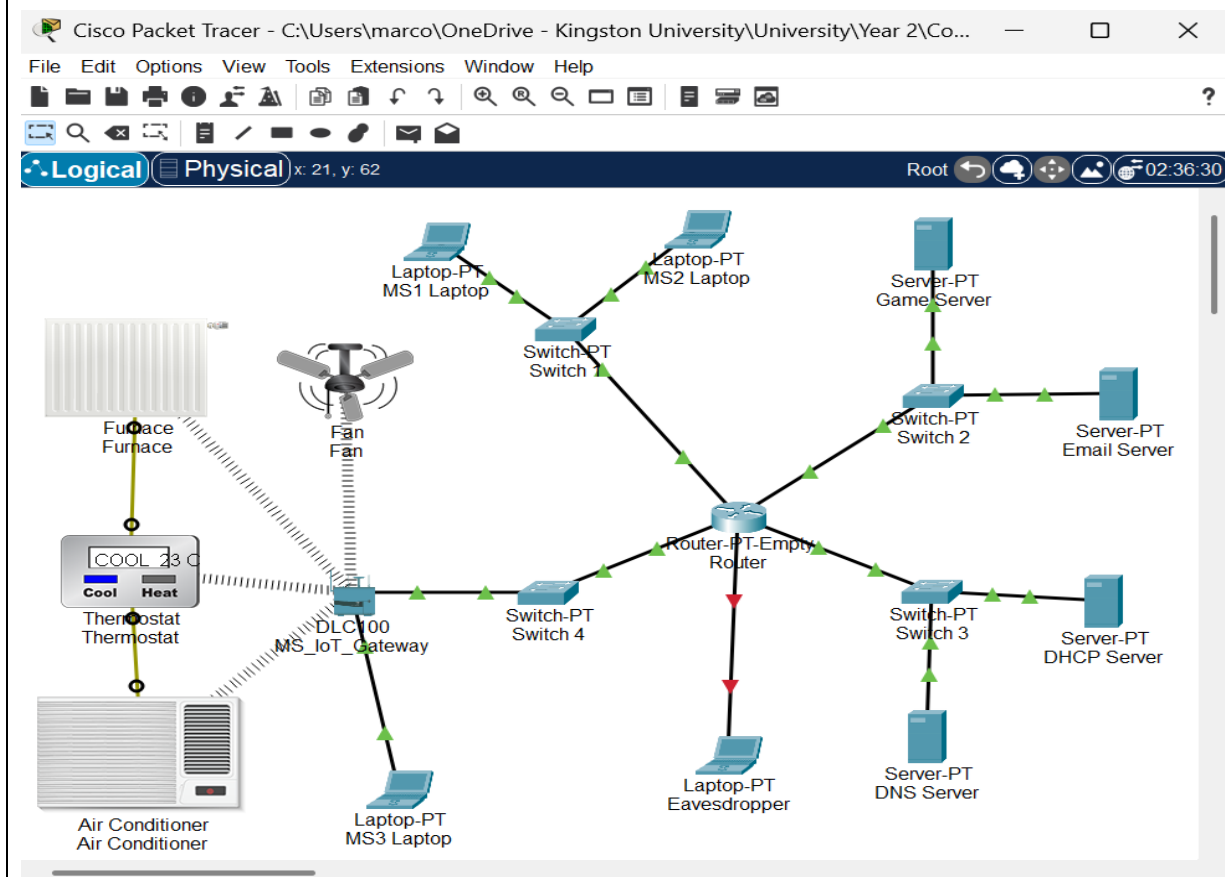
Actions	Enabled	Name	Condition	Actions
Edit Remove	Yes	Heating	Thermostat Temperature <= 16.0 °C	Set Furnace On to true Set Air Conditioner On to false Set Fan Status to Off Set Thermostat Status to Heating
Edit Remove	Yes	Cooling	Thermostat Temperature >= 23.0 °C	Set Furnace On to false Set Air Conditioner On to true Set Fan Status to High Set Thermostat Status to Cooling

[Add](#)

Snapshot 1: The settings to implement the two rules for the smart home system.



Snapshot 2: The Furnace turns on when the temperate crosses the threshold.



Snapshot 3: The Air Conditioner turns on when the temperate crosses the threshold.

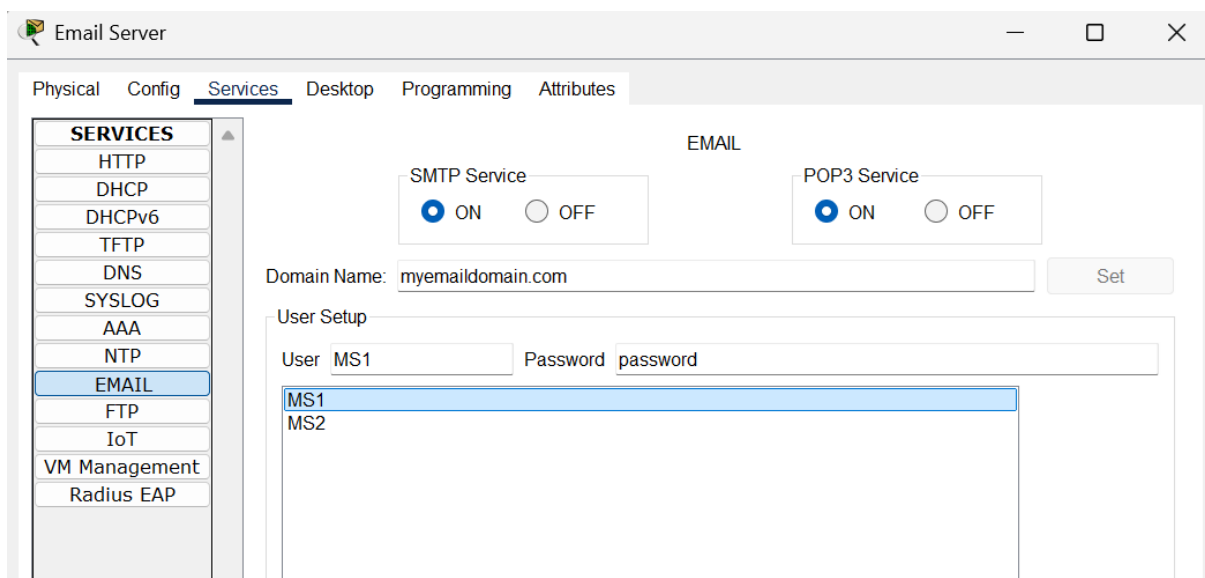
PICK ONLY ONE TASK FROM THE FOLLOWING THREE AND DELETE THE OTHER TWO AS APPROPRIATE.

Task 6

1. Using bullet points (max 5), give step-by-step implementation of the email exchange server. Use the knowledge you gained from the lectures, do not provide any vague answers (Word limit: 100)

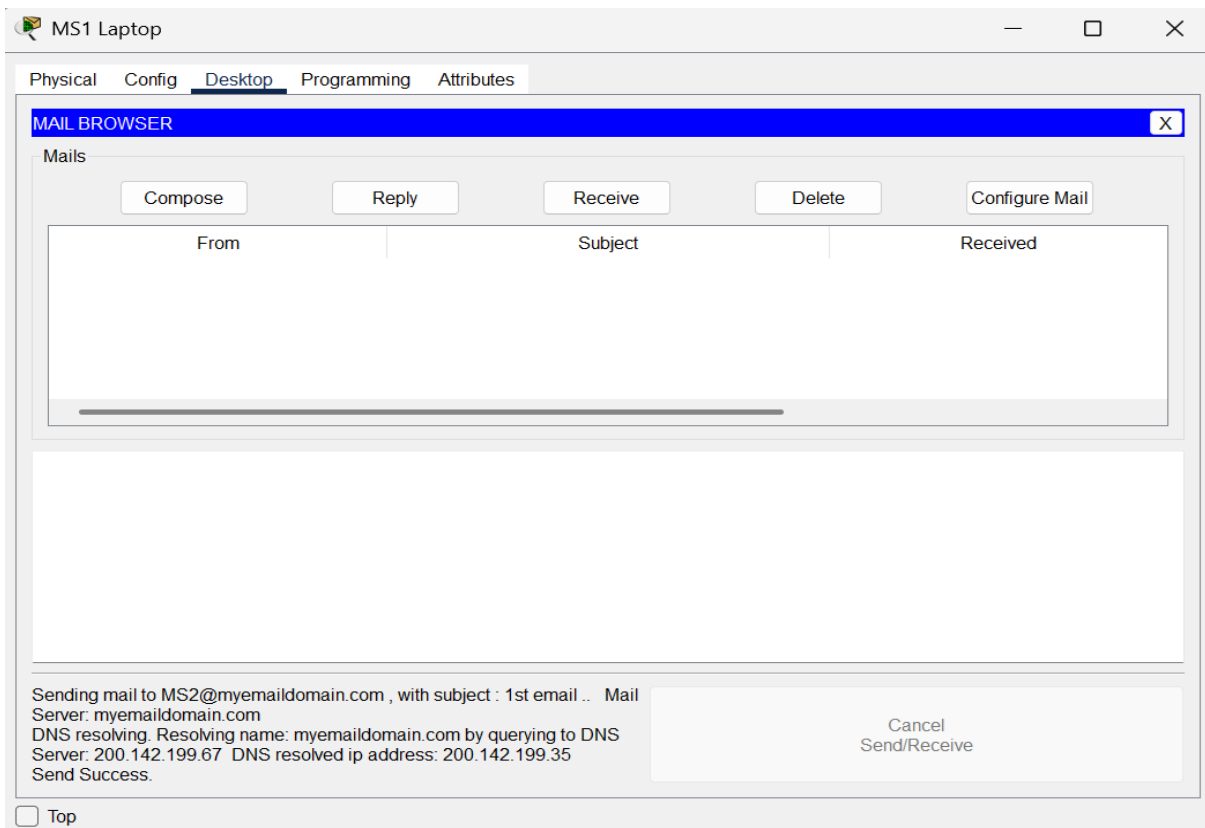
1. Add a DNS record on the DNS Server for the Email Server:
 - Click on the DNS Server > Services > DNS, add a name record (myemaildomain.com), of the type (A Record) and enter the address of the Email Server (200.142.199.35) then click add.
2. Configure the email server domain name:
 - Click on the Email Server > Services > Email check if the SMTP and POP3 services are “on”, enter the domain name (myemaildomain.com) and press set.
3. Configure email on laptops MS1 and MS2:
 - Click on the laptop > Desktop > Email and enter the name for that email account (example: MS1), the email address for that account (example: MS1@myemaildomain.com), enter the incoming and outgoing mail server (myemaildomain.com), enter logon information user name (example: MS1), and password (“password”) replace what’s within quotes with your password, and click save.
4. Add user email accounts to the Email Server:
 - Go back to the Email Server > Services > Email, in the user setup enter all the user names for the accounts created in the laptops (user MS1, password) add to the list (user MS2, password) add to the list.

2. Provide one evidence for the implementation of the Email Server (Readable SNAPSHOT).

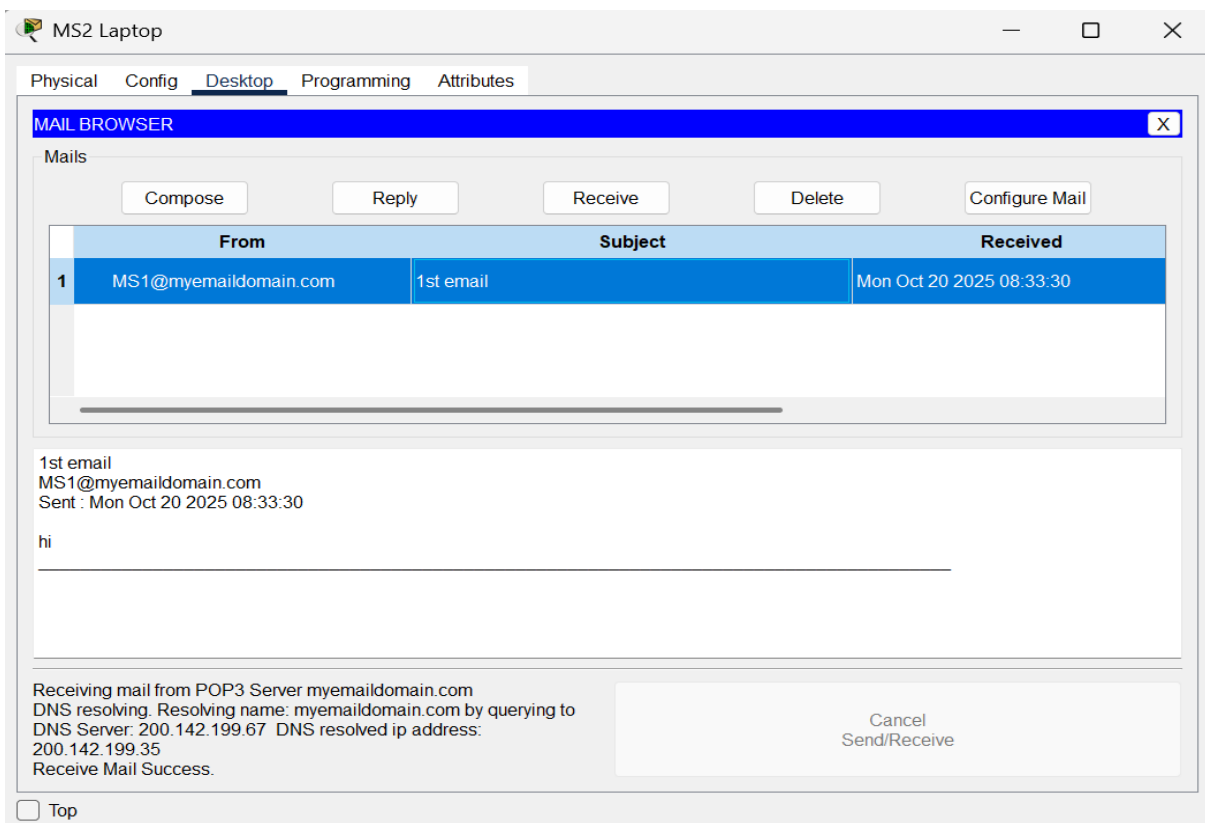


Snapshot 1: Configuration settings for the email exchange server.

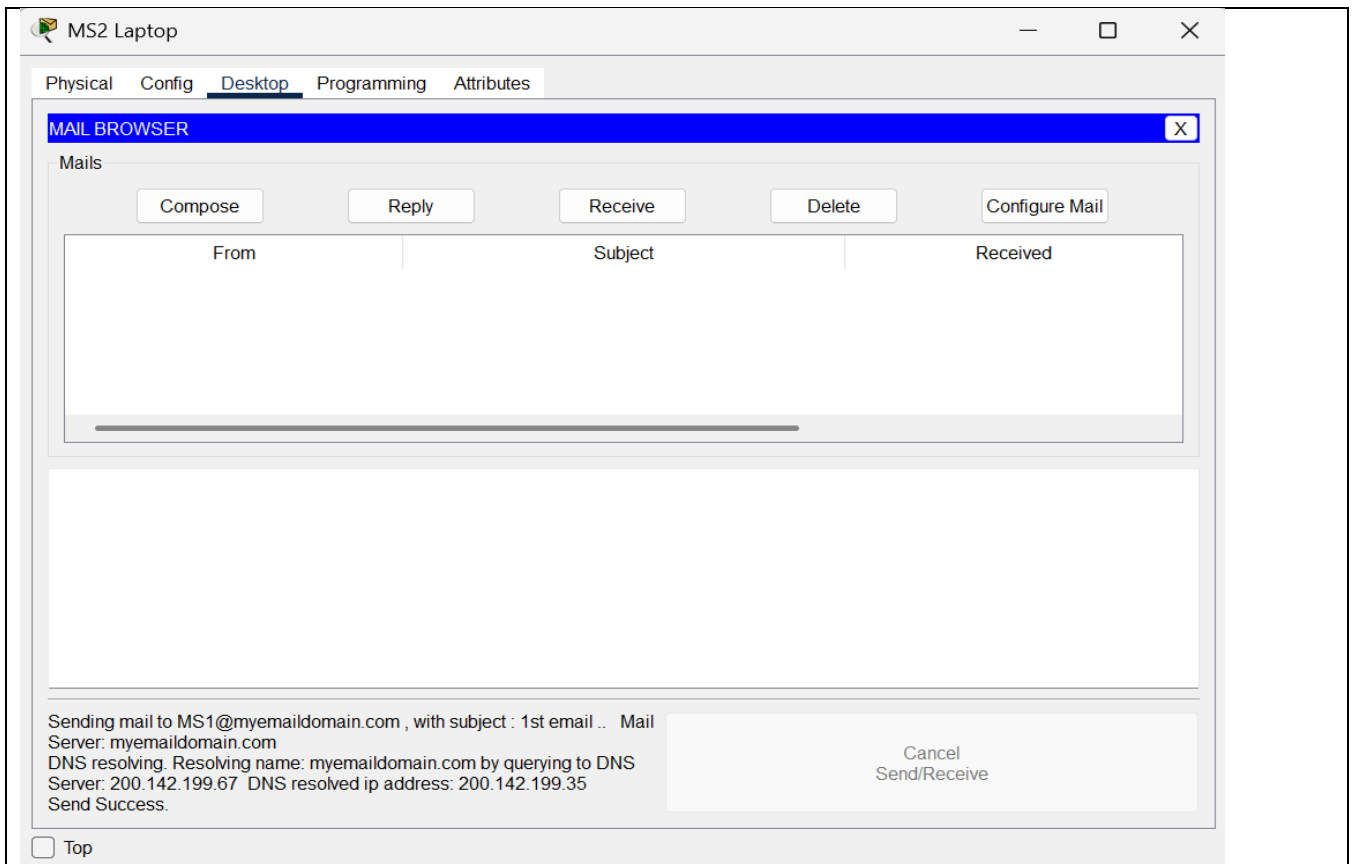
3. Provide a snapshot for the successful email exchange between the two laptops in subnet 1 (Readable SNAPSHOTS).



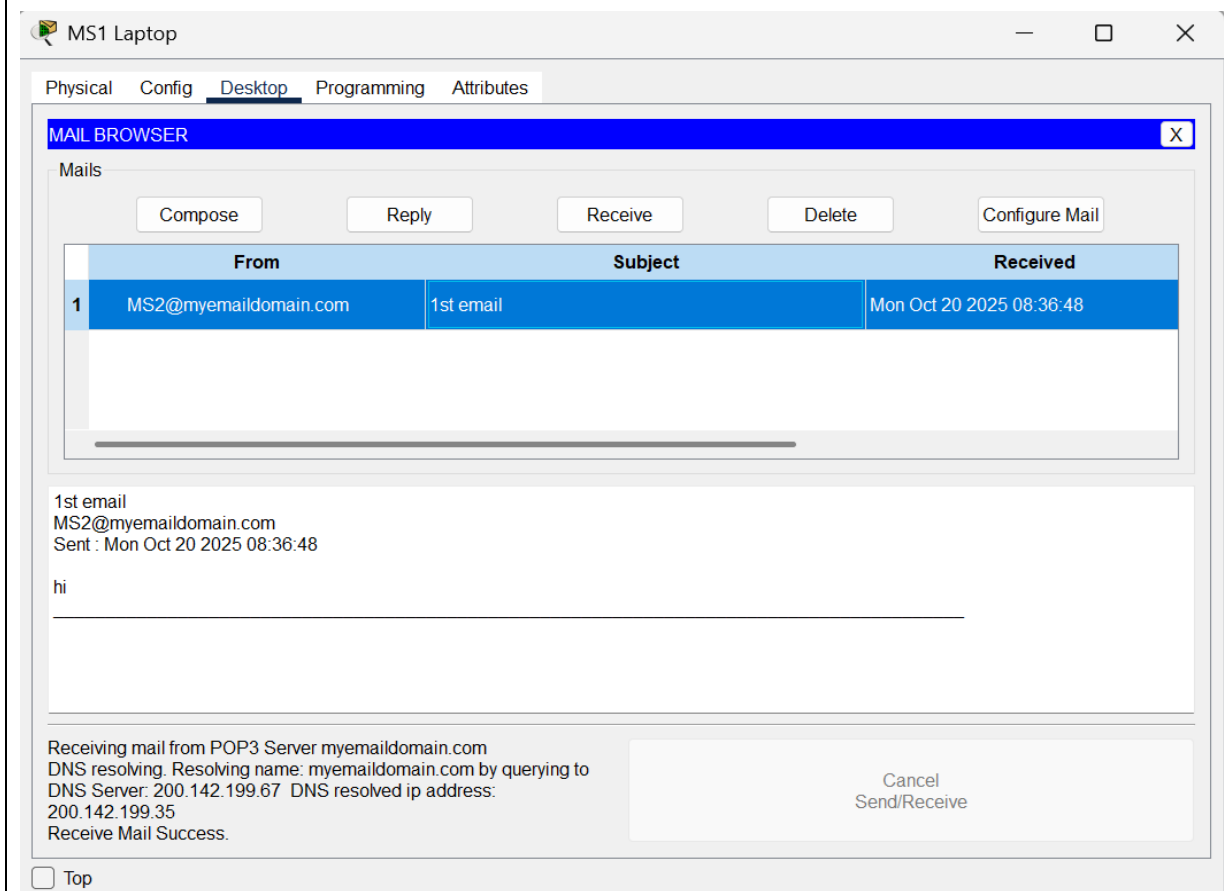
Snapshot 1: Email sent from Laptop 1



Snapshot 2: Email received at Laptop 2 from Laptop 1



Snapshot 3: Email sent from Laptop 2



Snapshot 4: Email received at Laptop 1 from Laptop 2

Acknowledgement of GAI Contribution

Did you use GAI? Yes/No

(If yes, then please fill the following as appropriate, in case of no, just add your personal information at the bottom of this page and leave the rest blank)

What GAI Tool(s) did you use? ... *for example, ChatGPT*

In the following table, provide all the prompts, in the right order, you used with GAI to understand the solution per task. You can provide multiple prompts for a single task if necessary. Feel free to add more rows to this table.

<i>Task Number</i>	<i>Prompts used to discuss the task with GAI</i>	<i>Did it provide the right answer?</i>

Explain in a short paragraph (5-8 lines), how useful was GAI in completing all the tasks for this coursework?

Provide the following information:

Name: Marco Pereira da Silva KU ID: K2432045 Date: 18/11/2025